

LAB 3

Motion sensor

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OUTLINE

01 Prelab

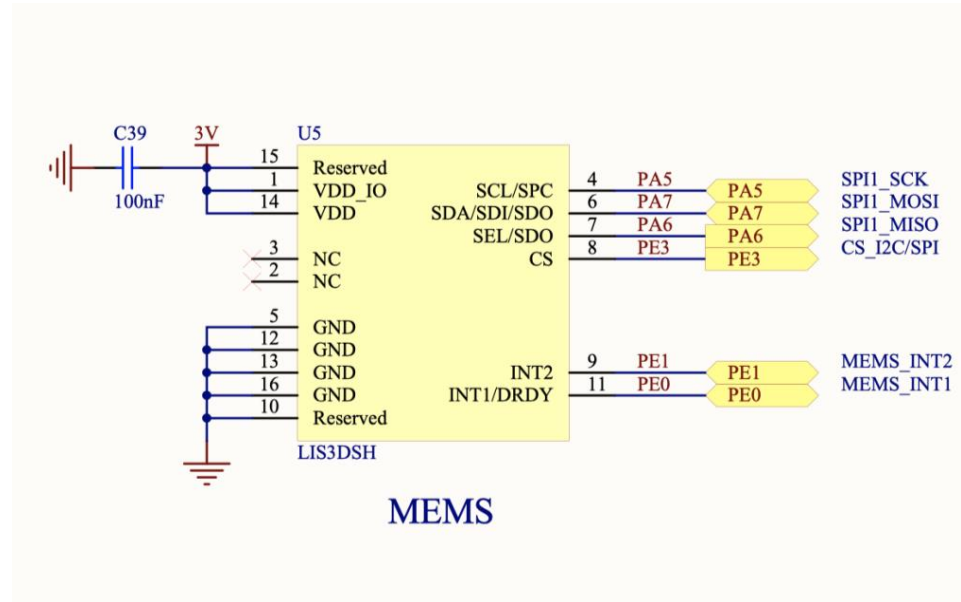
02 Lab 3 Requirement

03 Semaphore Introduction

04 Tools Introduction

Motion Sensor

- Lab 預設 motion sensor 型號: ST MEMS LIS3DSH
- The STM32F407VG microcontroller controls this motion sensor through the SPI interface.



Motion Sensor Setup

The screenshot displays the STM32CubeMX software interface, which is used for configuring microcontroller projects. The interface is divided into several sections:

- Pinout & Configuration:** This section on the left contains a search bar and a list of categories. Under the 'Connectivity' category, the 'SPI1' peripheral is highlighted with a red arrow labeled '1'.
- Software Packs:** This section in the top middle shows the 'SPI1 Mode and Configuration' settings. The 'Mode' is set to 'Full-Duplex Master' and the 'Hardware NSS Signal' is set to 'Disable'. A red arrow labeled '2' points to this configuration area.
- Pinout:** This section on the right shows a pinout view of the STM32F407VGTx microcontroller. The 'SPI1' pin is highlighted with a red arrow labeled '3'.

The microcontroller chip shown is an STM32F407VGTx LQFP100 package. The pinout view shows the various pins and their functions, including SPI1, I2C1, I2C2, I2C3, I2C4, I2C5, I2C6, I2C7, I2C8, I2C9, I2C10, I2C11, I2C12, I2C13, I2C14, I2C15, I2C16, I2C17, I2C18, I2C19, I2C20, I2C21, I2C22, I2C23, I2C24, I2C25, I2C26, I2C27, I2C28, I2C29, I2C30, I2C31, I2C32, I2C33, I2C34, I2C35, I2C36, I2C37, I2C38, I2C39, I2C40, I2C41, I2C42, I2C43, I2C44, I2C45, I2C46, I2C47, I2C48, I2C49, I2C50, I2C51, I2C52, I2C53, I2C54, I2C55, I2C56, I2C57, I2C58, I2C59, I2C60, I2C61, I2C62, I2C63, I2C64, I2C65, I2C66, I2C67, I2C68, I2C69, I2C70, I2C71, I2C72, I2C73, I2C74, I2C75, I2C76, I2C77, I2C78, I2C79, I2C80, I2C81, I2C82, I2C83, I2C84, I2C85, I2C86, I2C87, I2C88, I2C89, I2C90, I2C91, I2C92, I2C93, I2C94, I2C95, I2C96, I2C97, I2C98, I2C99, I2C100.

Motion Sensor Setup

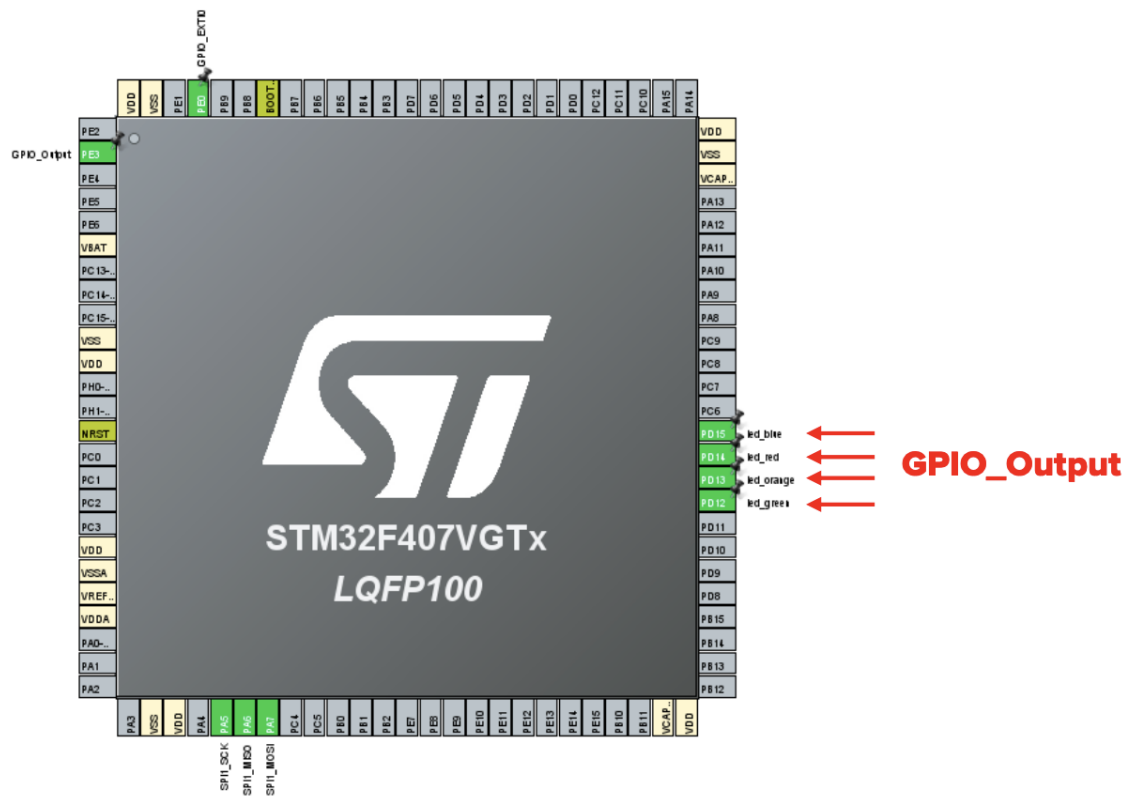
The image displays the STM32CubeMX software interface for configuring a motion sensor setup. The left sidebar shows the 'Categories' list with 'SPI1' selected under 'Connectivity'. The main panel shows the 'SPI1 Mode and Configuration' settings, including 'Mode: Full-Duplex Master' and 'Hardware NSS Signal: Disable'. The 'Configuration' tab is active, showing 'Parameter Settings' for SPI1. The 'Basic Parameters' section shows 'Frame Format: Motorola', 'Data Size: 8 Bits', and 'First Bit: MSB First'. The 'Clock Parameters' section shows 'Prescaler (for Baud Rat...): 2', 'Baud Rate: 8.0 Mbits/s', 'Clock Polarity (CPOL): Low', and 'Clock Phase (CPHA): 1 Edge'. The 'Advanced Parameters' section shows 'CRC Calculation: Disabled' and 'NSS Signal Type: Software'.

Red arrows point to specific pins on the pinout diagram:

- PE3 : GPIO_Output (for CS)** - Points to pin PE3.
- PE0 : GPIO_EXTIO (for INT)** - Points to pin PE0.

The pinout diagram shows the STM32F407VGTx LQFP100 package with pins labeled from PE0 to PE12, PA0 to PA12, and PB0 to PB12. The package is labeled 'STM32F407VGTx LQFP100'.

承接 lab0，記得設定 LED 燈相關 pin 腳



Motion Sensor Register

- 這些是能用來控制加速規的一些 register，事先 define 好每個的地址方便之後使用。

```
/* Private macro -----  
/* USER CODE BEGIN PM */  
#define LIS3DSH_WHO_AM_I_ADDR      0x0F  
#define LIS3DSH_STAT_ADDR          0x18  
#define LIS3DSH_CTRL_REG4_ADDR     0x20  
#define LIS3DSH_CTRL_REG1_ADDR     0x21  
#define LIS3DSH_CTRL_REG2_ADDR     0x22  
#define LIS3DSH_CTRL_REG3_ADDR     0x23  
#define LIS3DSH_CTRL_REG5_ADDR     0x24  
#define LIS3DSH_CTRL_REG6_ADDR     0x25  
  
#define LIS3DSH_STATUS_ADDR         0x27  
  
#define LIS3DSH_OUT_X_L_ADDR        0x28  
#define LIS3DSH_OUT_X_H_ADDR        0x29  
#define LIS3DSH_OUT_Y_L_ADDR        0x2A  
#define LIS3DSH_OUT_Y_H_ADDR        0x2B  
#define LIS3DSH_OUT_Z_L_ADDR        0x2C  
#define LIS3DSH_OUT_Z_H_ADDR        0x2D  
  
#define LIS3DSH_ST1_1_ADDR          0x40  
#define LIS3DSH_ST1_2_ADDR          0x41  
#define LIS3DSH_THRS1_1_ADDR        0x57  
#define LIS3DSH_MASK1_B_ADDR        0x59  
#define LIS3DSH_MASK1_A_ADDR        0x5A  
#define LIS3DSH_SETT1_ADDR          0x5B  
/* USER CODE END PM */
```

MEMS_Read / MEMS_Write

- 透過 SPI 來讀寫 register 的函式。

```
/* USER CODE BEGIN 0 */  
- void MEMS_Write(uint8_t address, uint8_t data){  
    HAL_GPIO_WritePin(GPIOE, GPIO_PIN_3, GPIO_PIN_RESET);  
    HAL_SPI_Transmit(&hspi1, &address, 1, 10);  
    HAL_SPI_Transmit(&hspi1, &data, 1, 10);  
    HAL_GPIO_WritePin(GPIOE, GPIO_PIN_3, GPIO_PIN_SET);  
}  
- void MEMS_Read(uint8_t address, uint8_t *data){  
    address |= 0x80;  
    HAL_GPIO_WritePin(GPIOE, GPIO_PIN_3, GPIO_PIN_RESET);  
    HAL_SPI_Transmit(&hspi1, &address, 1, 10);  
    HAL_SPI_Receive(&hspi1, data, 1, 10);  
    HAL_GPIO_WritePin(GPIOE, GPIO_PIN_3, GPIO_PIN_SET);  
}  
/* USER CODE END 0 */
```

Motion Sensor Testing

- 透過讀取 **WHO_AM_I register** 的值，來使綠燈閃爍。

```
97 void LED_Task(void *pvParameter)
98 {
99     for(;;){
100         uint8_t data;
101         MEMS_Read(LIS3DSH_WHO_AM_I_ADDR,&data);
102         if(data == 0x3F){
103             HAL_GPIO_TogglePin(GPIOD, GPIO_PIN_12);
104         }
105         vTaskDelay(500/portTICK_RATE_MS);
106     }
107 }
108 }
```

‡ **WHO_AM_I (0Fh)**

Who_AM_I register.

Table 20. WHO_AM_I register default values

0	0	1	1	1	1	1	1
---	---	---	---	---	---	---	---

預設值0x3F

[測試 code 連結](#)

已包括 6、7、8 頁的 code

理論上如果沒有問題，板子的綠色 LED 燈會閃爍
(承接 lab0，記得設定綠色 LED 燈相關 pin 腳)

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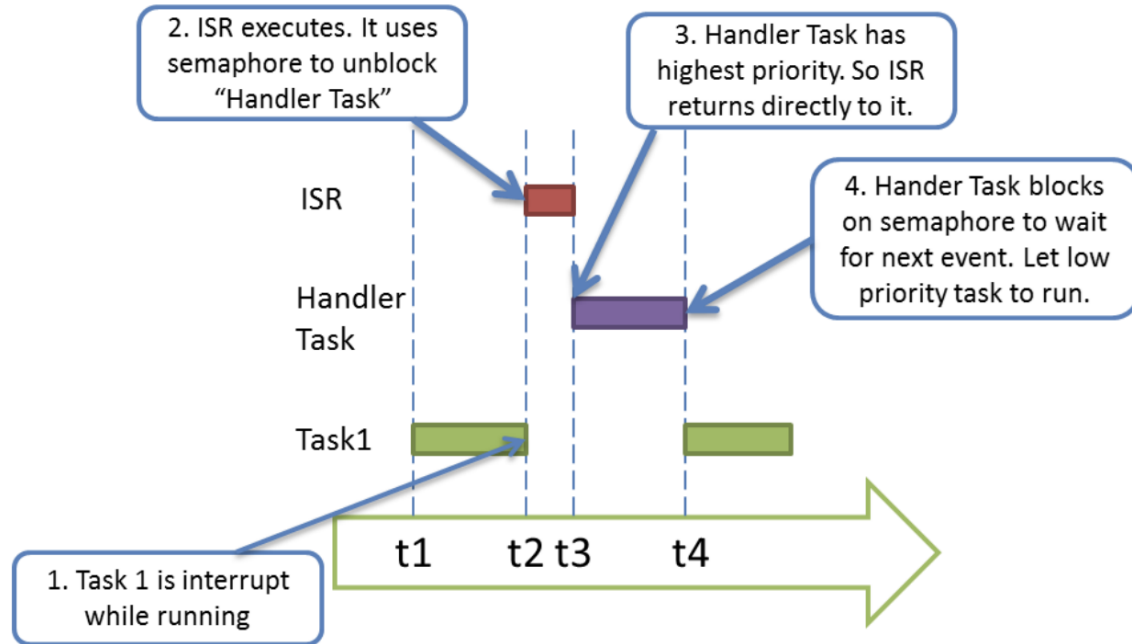
03 Semaphore Introduction

04 Tools Introduction

Lab 3 Introduction

- Must use sensor interrupt : motion detection.
 - When you shake your board, it will trigger the interrupt.
- Please use the deferred interrupt handling task.
 - Use semaphore.
 - ISR will give the semaphore and the handler task enters the running state.
- At the beginning, the **green LED blinking**, then shake the board, the **red LED triggered** (switch state) by ISR and the **orange LED blinking five times** in handler task.
- When orange LED blinking, you should not trigger the sensor interrupt if you shake the board.

Deferred Interrupt Handling



Lab 3 Grading (8%)

- (3%) 正確觸發 interrupt，且執行 ISR (即搖晃板子有用)
- (2%) 不會無緣無故解開鎖執行 handler task，
且在 handler task 執行完之前無法觸發 interrupt。
- (3%) Lab report

[DEMO 影片參考](#)

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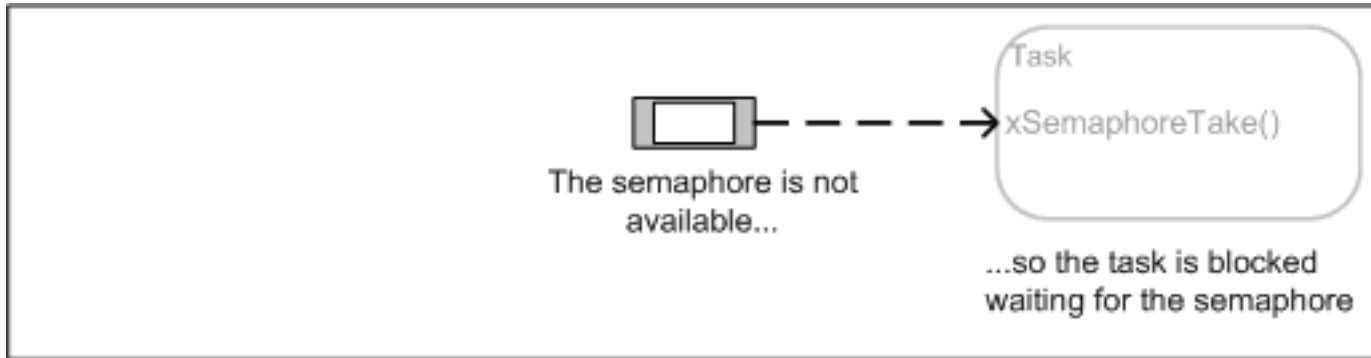
04 Tools Introduction

FreeRTOS Semaphore

- Three types of semaphores
 - Binary
 - Counting
 - Mutex
- Often used to control access to shared resources and synchronization

Binary Semaphores

- Can be used for synchronization
- Do not support priority inheritance protocol
- Binary semaphores can be used in ISRs

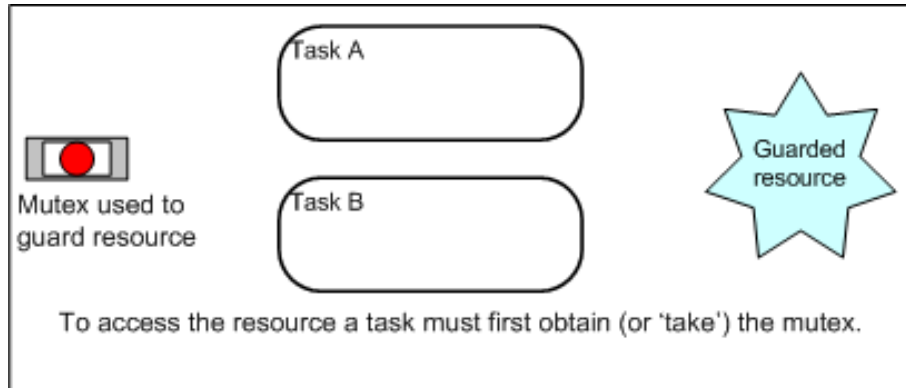


Counting Semaphores

- Counting pending events
 - An event handler (e.g. ISR) gives a semaphore S each time an event occurs ($S.value++$)
 - A handler task takes S each time it processes an event ($S.value--$)
 - When the semaphore S is created, $S.value = 0$
- Resource management
 - Get a resource ($S.value--$)
 - Release a resource ($S.value++$)
 - When the semaphore S is created, $S.value = N$ (resource)

Mutexes

- Mutexes are binary semaphores that include a priority inheritance mechanism.
- A better choice for implementing simple mutual exclusion
- Mutexes should NOT be used from an interrupt
 - priority inheritance only makes sense for tasks, not ISRs
 - An ISR should not block to wait for a resource



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API Reference (#include "semphr.h")

- [xSemaphoreCreateBinary](#)
- [xSemaphoreGiveFromISR](#)
- [xSemaphoreTake](#)

ISR

- Drivers/STM32F4xx_HAL_Driver/Src/stm32f4xx_hal_gpio.c

```
/**
 * @brief This function handles EXTI interrupt request.
 * @param GPIO_Pin Specifies the pins connected EXTI line
 * @retval None
 */
void HAL_GPIO_EXTI_IRQHandler(uint16_t GPIO_Pin)
{
    /* EXTI line interrupt detected */
    if(__HAL_GPIO_EXTI_GET_IT(GPIO_Pin) != RESET)
    {
        HAL_GPIO_EXTI_CLEAR_IT(GPIO_Pin);
        HAL_GPIO_EXTI_Callback(GPIO_Pin);
    }
}

/**
 * @brief EXTI line detection callbacks.
 * @param GPIO_Pin Specifies the pins connected EXTI line
 * @retval None
 */
weak void HAL_GPIO_EXTI_Callback(uint16_t GPIO_Pin)
{
    /* Prevent unused argument(s) compilation warning */
    UNUSED(GPIO_Pin);
    /* NOTE: This function should not be modified, when the callback is needed,
    the HAL_GPIO_EXTI_Callback could be implemented in the user file */
}
```

- You can define a new function “void HAL_GPIO_EXTI_Callback(uint16_t GPIO_Pin)” in main.c (For ISR)

Motion Sensor

- SM1_EN : 1

7.21 CTRL_REG1 (21h)

SM1 control register.

Table 56. SM1 control register

HYST2_1	HYST1_1	HYST0_1	-	SM1_PIN	-	-	SM1_EN
---------	---------	---------	---	---------	---	---	--------

Table 57. SM1 control register structure

HYST2_1 HYST1_1 HYST0_1	Hysteresis unsigned value to be added or subtracted from threshold value in SM1 Default value: 000
SM1_PIN	0: SM1 interrupt routed to INT1; 1: SM1 interrupt routed to INT2 pin Default value: 0
SM1_EN	0: SM1 disabled; 1: SM1 enabled Default value: 0

Motion Sensor

- IEA : 1
- INT1_EN : 1

7.23 CTRL_REG3 (23h)

Control register 3.

Table 60. Control register 3

DR_EN	IEA	IEL	INT2_EN	INT1_EN	VFILT	-	STRT
-------	-----	-----	---------	---------	-------	---	------

Table 61. CTRL_REG3 register description

DR_EN	DRDY signal enable to INT1. Default value: 0 (0: data ready signal not connected; 1: data ready signal connected to INT1)
IEA	Interrupt signal polarity. Default value: 0 (0: interrupt signals active LOW; 1: interrupt signals active HIGH)
IEL	Interrupt signal latching. Default value: 0 (0: interrupt signal latched; 1: interrupt signal pulsed)
INT2_EN	Interrupt 2 enable/disable. Default value: 0 (0: INT2 signal disabled; 1: INT2 signal enabled)
INT1_EN	Interrupt 1 enable/disable. Default value: 0 (0: INT1/DRDY signal disabled; 1: INT1/DRDY signal enabled)
VFILT	Vector filter enable/disable. Default value: 0 (0: vector filter disabled; 1: vector filter enabled)
STRT	Soft reset bit (0: no soft reset; 1: soft reset (POR function))

Motion Sensor

- ODR : 0110
 - 100 Hz
- Zen : 1
- Yen : 1
- Xen : 1

7.20 CTRL_REG4 (20h)

Control register 4.

Table 53. Control register 4

ODR3	ODR2	ODR1	ODR0	BDU	Zen	Yen	Xen
------	------	------	------	-----	-----	-----	-----

Table 54. CTRL_REG4 register description

ODR 3:0	Output data rate and power mode selection. Default value: 0000 (see Table 55)
BDU	Block data update. Default value: 0 (0: continuous update; 1: output registers not updated until MSB and LSB have been read)
Zen	Z-axis enable. Default value: 1 (0: Z-axis disabled; 1: Z-axis enabled)
Yen	Y-axis enable. Default value: 1 (0: Y-axis disabled; 1: Y-axis enabled)
Xen	X-axis enable. Default value: 1 (0: X-axis disabled; 1: X-axis enabled)

Motion Sensor

7.38 THRS1_1 (57h)

Threshold value for SM1 operation.

Table 100. THRS1_1 register

THS7	THS6	THS5	THS4	THS3	THS2	THS1	THS0
------	------	------	------	------	------	------	------

Table 101. THRS1_1 register description

THS[7:0]	Threshold values for SM1. Default value: 0000 0000
----------	--

Motion Sensor

- P_X、N_X、P_Y、N_Y、P_Z、N_Z : 1

7.39 MASK1_B (59h)

Axis and sign mask (swap) for SM1 motion-detection operation.

Table 102. MASK1_B axis and sign mask register

P_X	N_X	P_Y	N_Y	P_Z	N_Z	P_V	N_V
-----	-----	-----	-----	-----	-----	-----	-----

Table 103. MASK1_B register structure

P_X	0: X + disabled; 1: X + enabled
N_X	0: X - disabled; 1: X - enabled
P_Y	0: Y+ disabled; 1: Y + enabled
N_Y	0: Y- disabled; 1: Y - enabled
P_Z	0: Z + disabled; 1: Z + enabled
N_Z	0: Z - disabled; 1: Z - enabled
P_V	0: V + disabled; 1: V + enabled
N_V	0: V - disabled; 1: V - enabled

Motion Sensor

- P_X、N_X、P_Y、N_Y、P_Z、N_Z : 1

7.40 MASK1_A (5Ah)

Axis and sign mask (default) for SM1 motion-detection operation.

Table 104. MASK1_A axis and sign mask register

P_X	N_X	P_Y	N_Y	P_Z	N_Z	P_V	N_V
-----	-----	-----	-----	-----	-----	-----	-----

Table 105. MASK1_A register structure

P_X	0: X + disabled; 1: X + enabled
N_X	0: X - disabled; 1: X - enabled
P_Y	0: Y + disabled; 1: Y + enabled
N_Y	0: Y - disabled; 1: Y - enabled
P_Z	0: Z + disabled; 1: Z + enabled
N_Z	0: Z - disabled; 1: Z - enabled
P_V	0: V + disabled; 1: V + enabled
N_V	0: V - disabled; 1: V - enabled

Motion Sensor

- SITR : 1

7.41 SETT1 (5Bh)

Setting of threshold, peak detection and flags for SM1 motion-detection operation.

Table 106. SETT1 register structure

P_DET	THR3_SA	ABS	-	-	THR3_MA	R_TAM	SITR
-------	---------	-----	---	---	---------	-------	------

Table 107. SETT1 register description

P_DET	SM1 peak detection. Default value: 0 (0: peak detection disabled; 1: peak detection enabled)
THR3_SA	Default value: 0 (0: no action; 1: threshold 3 limit value for axis and sign mask reset (MASKB_1))
ABS	Default value: 0 (0: unsigned thresholds; 1: signed thresholds)
THR3_MA	Default value: 0 (0: no action; 1: threshold 3 limit value for axis and sign mask reset (MASKA_1))
R_TAM	Next condition validation flag. Default value: 0 (0: no valid next condition found; 1: valid next condition found and reset)
SITR	Default value: 0 (0: no actions; 1: program flow can be modified by STOP and CONT commands)

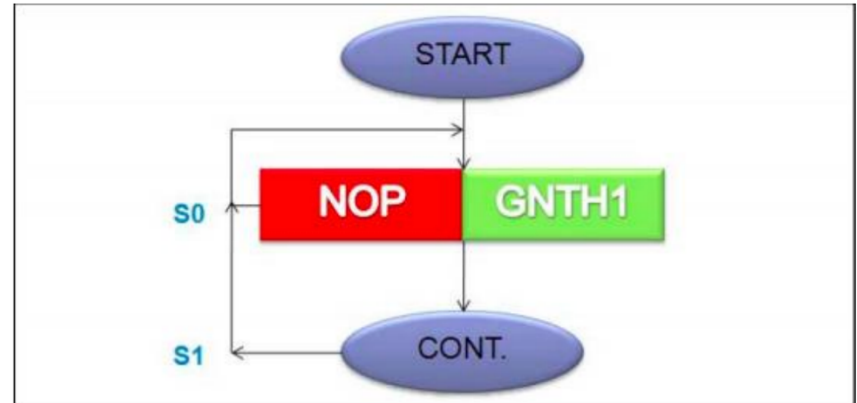
Motion Sensor

7.32 STx_1 (40h-4Fh)

State machine 1 code register STx_1 (x = 1-16).

State machine 1 system register is made up of 16, 8-bit registers to implement 16-step op-code.

ST1_1	40h	05h
-------	-----	-----



Motion Sensor

6.1.6 GNTH1 (5h)

The GNTH1 condition is valid if any/triggered axis of the data sample set (X, Y, Z, V) is greater than threshold 1 level.

Threshold is: THRS1_y + Hysteresis.

Hysteresis is:

- State Machine 1: CTRL_REG1, bits HYST2_1, HYST1_1 and HYST0_1;
- State Machine 2: CTRL_REG2, bits HYST2_2, HYST1_2 and HYST0_2.

This condition affects or is affected by the following registers:

- THRS1_y: Threshold 1 value;
- MASKy_A and MASKy_B: Axis mask filter values;
- SETTy, bit ABS: Unsigned/signed settings;
- SETTy, bit R_TAM: Release temporary output mask settings;
- SETTy, bit P_DET: Peak detection settings;
- PEAKy: Peak output value;
- PRy: Program and Reset pointer addresses.

Motion Sensor

6.2.2 CONT (11h)

The CONT command loops execution to the beginning.

This command has no parameters and it is an "Immediately executed" type.

Actions:

1. If SETTy, bit SITR = 1:
 - OUTSy is updated to selected temporary mask value;
 - Set output register (and signal if selected): STAT, bit INT_SMy = 1.
2. Default initial start executed
3. Continue execution from step address PPy = 0

This command affects or is affected by the following registers:

- State Machine y is enabled/disabled: CTRL_REG1, bit SM1_EN is set to 0/1 for State Machine 1. CTRL_REG2, bit SM2_EN is set to 0/1 for State Machine 2;
- SETTy, bit SITR: Defines output functionality of STOP command;
- OUTSy: Output value of State Machine y;
- STAT, bit INT_SMy: Indicator of valid interrupt action;
- PRy: Program and Reset pointer addresses.

Motion Sensor

OUTS1 (5Fh)

Output flags on axis for interrupt SM1 management.

Table 114. OUTS1 register

P_X	N_X	P_Y	N_Y	P_Z	N_Z	P_V	N_V
-----	-----	-----	-----	-----	-----	-----	-----

Read action of this register, depending on the flag affects SM1 interrupt functions.

Table 115. OUTS1 register description

P_X	0: X + no show; 1: X+ show
N_X	0: X - no show; 1: X – show
P_Y	0: Y + no show; 1: Y + show
N_Y	0: Y - no show; 1: Y – show
P_Z	0: Z + no show; 1: Z + show
N_Z	0: Z - no show; 1: Z – show
P_V	0: V + no show; 1: V + show
N_V	0: V - no show; 1: V – show

HINT

- Handler task must have a higher priority.
- `HAL_NVIC_SetPriority(EXTIO_IRQn, configLIBRARY_MAX_SYSCALL_INTERRUPT_PRIORITY, 0);`
- `HAL_NVIC_EnableIRQ(EXTIO_IRQn);`
- Read OUTS1 (5Fh) **after the handler task finishes** to clear the sensor interrupt status; otherwise, the motion interrupt may only be triggered once.
- [Template Code](#)